Straight Line: Formula Sheet

Geometry facts

Quadrilateral: $\frac{1}{2}d_1 d_2 \sin\theta$

Parallelogram:

$$ab\sin\theta = p_1 p_2 \operatorname{cosec}\theta$$

Diagonals: $2(a^2+b^2) = d_1^2 + d_2^2$

Trapezium: $\frac{1}{2}(a+b)h$, median $\frac{a+b}{2}$

Ptolemy: $AB \cdot CD + AD \cdot BC = AC \cdot BD$

Chords: $PA \cdot PB = PC \cdot PD = PT^2$

Points

Distance: $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

Polar: $x = r\cos\theta,\ y = r\sin\theta$

Rotation:

$$(x\cos\alpha - y\sin\alpha,\ x\sin\alpha + y\cos\alpha)$$

Internal:

$$\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$$

External:

$$\left(\frac{mx_2-nx_1}{m-n}, \frac{my_2-ny_1}{m-n}\right)$$

Harmonic:

$$\frac{2}{AB} = \frac{1}{AP} + \frac{1}{AQ}$$

Triangle centres

Centroid:

$$\left(\frac{\sum x_i}{3}, \frac{\sum y_i}{3}\right)$$

Incentre:

$$\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}\right)$$

Excentre I_1:

$$\left(\frac{-ax_1+bx_2+cx_3}{-a+b+c}, \frac{-ay_1+by_2+cy_3}{-a+b+c}\right)$$

Circumcentre: weights $\sin 2A, \sin 2B, \sin 2C$

Orthocentre: weights $\tan A, \tan B, \tan C$

Euler: $G = \dfrac{H+2O}{3}$; O at origin: $H = 3G$

4th vertex: $D = A + C - B$

Area

Triangle:

$$\frac{1}{2}|x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)|$$

Polygon:

$$\frac{1}{2}|\sum(x_i y_{i+1} - x_{i+1}y_i)|$$

Three lines:

$$\frac{(\det[a_i\ b_i\ c_i])^2}{2|C_1C_2C_3|}$$

Equilateral: $\dfrac{h^2}{\sqrt{3}}$; right isosceles: h^2

Parallelogram (lines):

$$\frac{|(c_1-d_1)(c_2-d_2)|}{|a_1b_2 - a_2b_1|}$$

Line through (a,b): least area $2ab$, least $OA+OB = (\sqrt{a}+\sqrt{b})^2$

Equation of a line

Slope:

$$m = \tan\theta = \frac{y_2-y_1}{x_2-x_1} = -\frac{a}{b}$$

Point-slope: $y - y_1 = m(x-x_1)$

Slope-intercept: $y = mx + c$

Intercept:

$$\frac{x}{a} + \frac{y}{b} = 1$$

Normal: $x\cos\alpha + y\sin\alpha = p$

Parametric:

$$(x_1 + r\cos\theta,\ y_1 + r\sin\theta)$$

Two-point:

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Equal intercepts: $m = -1$; equally inclined: $m = \pm 1$

Angles and distances

Angle:

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

Parallel: $m_1 = m_2$; perpendicular: $m_1 m_2 = -1$

Point to line:

$$\frac{|ax_1 + by_1 + c|}{\sqrt{a^2+b^2}}$$

Parallel lines:

$$\frac{|c_1 - c_2|}{\sqrt{a^2+b^2}}$$

Foot:

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = -\frac{ax_1+by_1+c}{a^2+b^2}$$

Image:

$$\frac{h-x_1}{a} = \frac{k-y_1}{b} = -\frac{2(ax_1+by_1+c)}{a^2+b^2}$$

Position, concurrency, images

Same side: $L(P)\,L(Q) > 0$

Concurrent:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

Images of (a, b):

$$(a, -b), (-a, b), (-a, -b), (b, a), (-b, -a)$$

Family: $L_1 + \lambda L_2 = 0$

Diagonals: $\dfrac{u}{k} + \dfrac{v}{l} = 1,\ \dfrac{u}{k} = \dfrac{v}{l}$

Transformations and bisectors

Shift: $x = X + h,\ y = Y + k$

Rotate axes:

$$x = X\cos\theta - Y\sin\theta,\ y = X\sin\theta + Y\cos\theta$$

Bisectors:

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

Origin bisector $(c_1, c_2 > 0)$: acute if $a_1a_2 + b_1b_2 < 0$

Triangle: $BD : DC = AB : AC$

Pair of lines

Sum, product:

$$-\frac{2h}{b},\ \frac{a}{b}$$

Angle:

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$$

Perpendicular $a + b = 0$; parallel $h^2 = ab$

Bisectors:

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

Pair condition: $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

Centre: $ax + hy + g = 0,\ hx + by + f = 0$

Parallel pair distance:

$$2\sqrt{\frac{g^2 - ac}{a(a + b)}}$$

Homogenise:

$$1 = \frac{lx + my}{n}$$